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DSTALGO

Name: Sonza, Zoilo Ynigo L.	Date: 10/14/2026
Section: BSIS02	Score:

1st Constant Time

```
using System;
using System.Windows.Forms;

namespace ConstantTime01
{
    public partial class Form1 : Form
    {
        // Sample array
        int[] numbers = { 10, 20, 30, 40, 50 };

        public Form1()
        {
            InitializeComponent();
        }

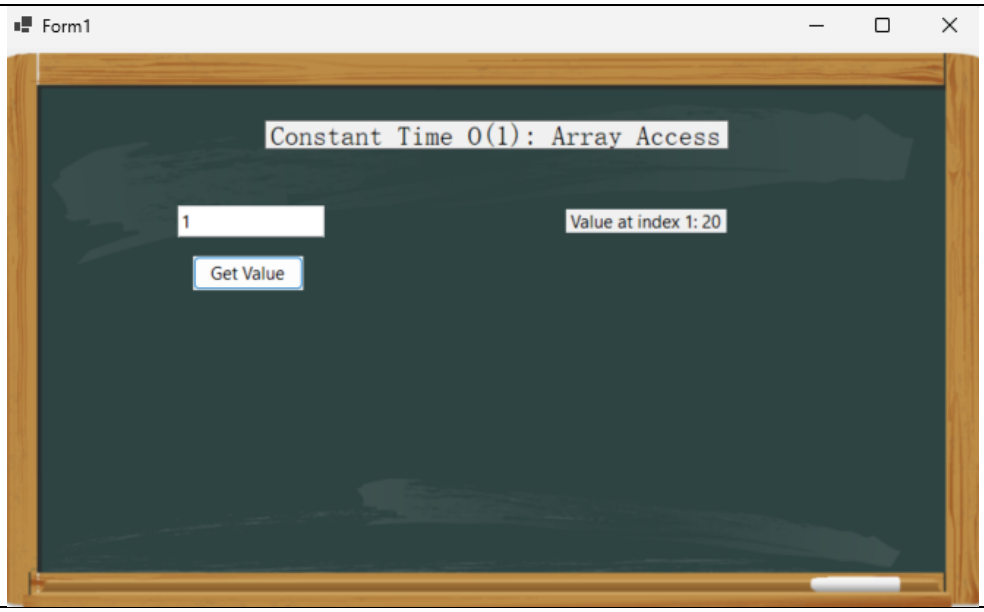
        private void btnAccess_Click(object sender, EventArgs e)
        {
            try
            {
                // Get index from textbox
                int index = int.Parse(txtIndex.Text);

                // Constant time access O(1)
                int answer = numbers[index];

                // Show result
                lblResult.Text = $"Value at index {index}: {answer}";
            }
            catch (Exception ex)
            {
                lblResult.Text = "Error: " + ex.Message;
            }
        }
    }
}
```

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2nd Constant Time

```
using System;
using System.Windows.Forms;

namespace ConstantAdd
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnAdd_Click(object sender, EventArgs e)
        {
            try
            {
                // Get numbers from textboxes
                int a = int.Parse(txtA.Text);
                int b = int.Parse(txtB.Text);

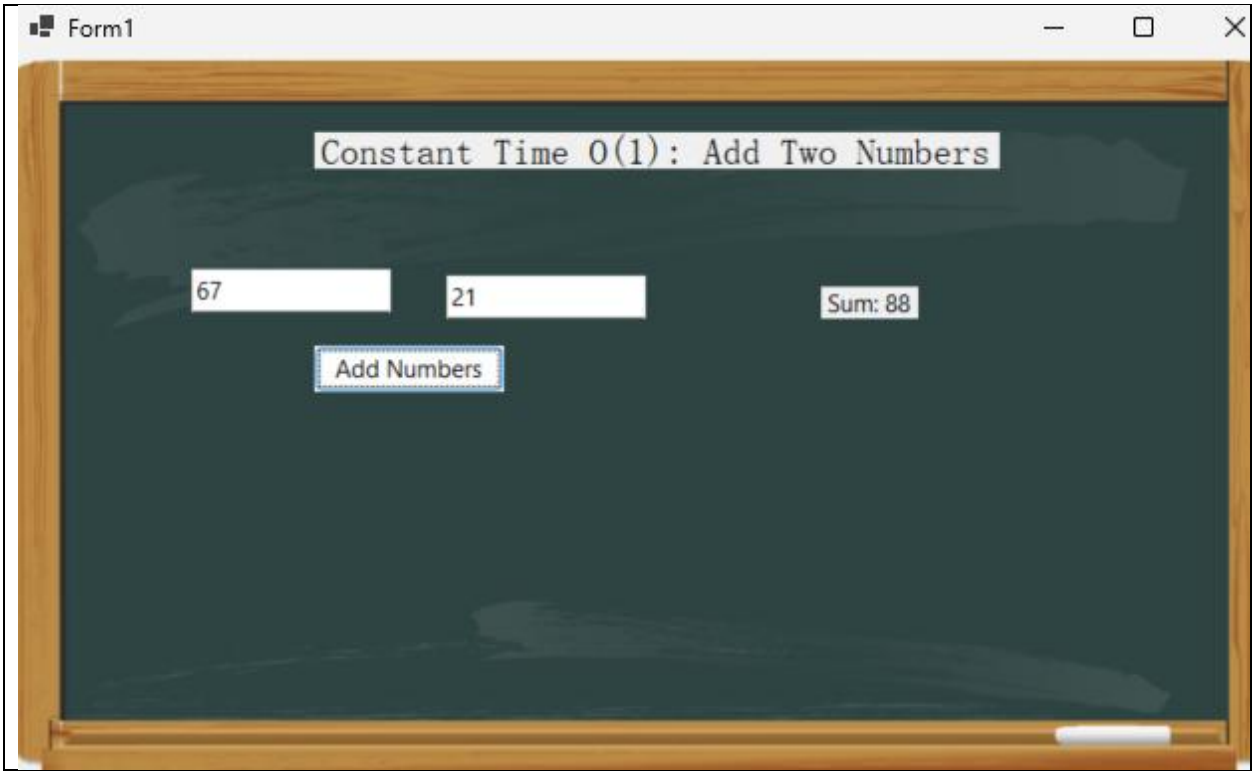
                // Constant time addition O(1)
                int sum = a + b;

                // Show result
                lblResult.Text = $"Sum: {sum}";
            }
            catch (Exception ex)
            {
                lblResult.Text = "Error: " + ex.Message;
```

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DSTALGO

```
            }
        }
    }
}
```



3rd Constant Time

```
using System;
using System.Windows.Forms;

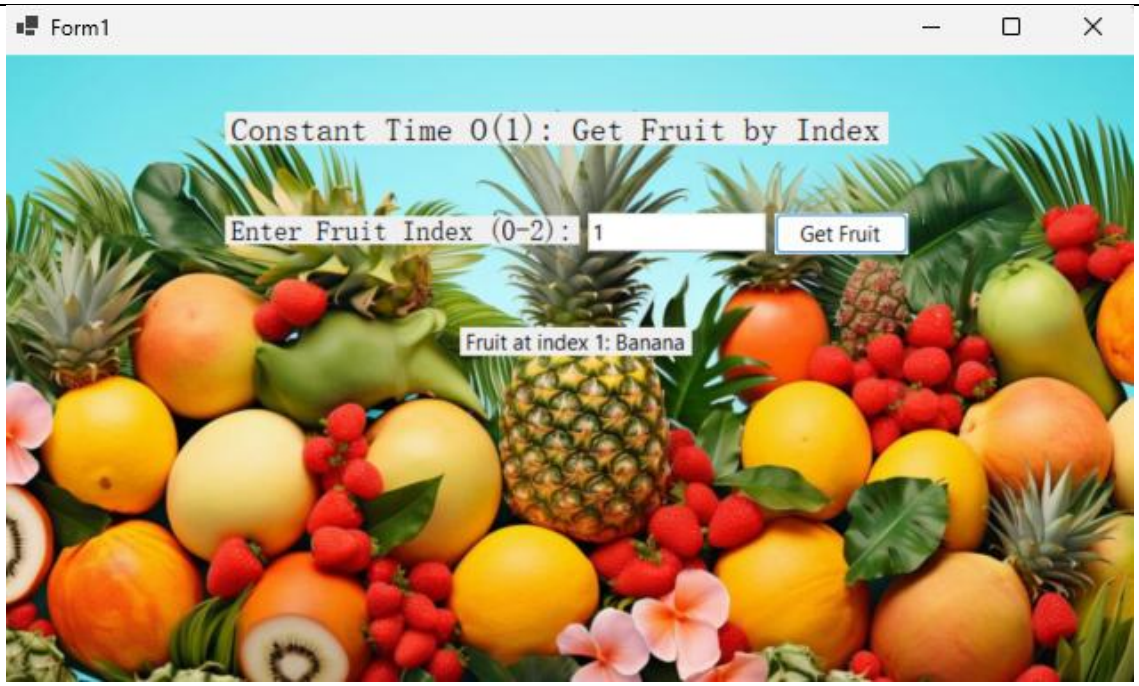
namespace GetFruit
{
    public partial class Form1 : Form
    {
        // Array of fruits
        string[] fruits = { "Apple", "Banana", "Mango" };

        public Form1()
        {
            InitializeComponent();
        }
```

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```
        private void btnGetFruit_Click(object sender, EventArgs e)
        {
            if (int.TryParse(txtIndex.Text.Trim(), out int index))
            {
                if (index >= 0 && index < fruits.Length)
                {
                    lblResult.Text = $"Fruit at index {index}: {fruits[index]}";
                }
                else
                {
                    lblResult.Text = "Index out of range. Please enter 0, 1, or 2.";
                }
            }
            else
            {
                lblResult.Text = "Invalid input.";
            }
        }
    }
}
```



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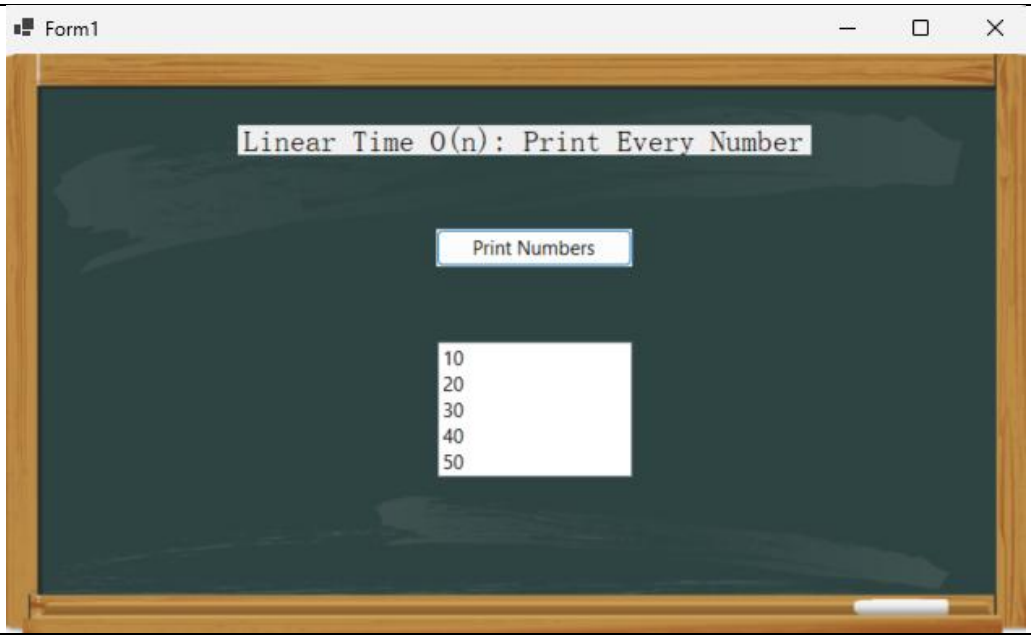
1st Linear Time

```
namespace PrintNumber
{
    public partial class Form1 : Form
    {
        // Array of numbers
        int[] numbers = { 10, 20, 30, 40, 50 };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnPrintNumbers_Click(object sender, EventArgs e)

        {
            // Clear previous output
            lstNumbers.Items.Clear();

            // Linear traversal O(n): print every number
            for (int i = 0; i < numbers.Length; i++)
            {
                lstNumbers.Items.Add(numbers[i]);
            }
        }
    }
}
```



2nd Linear Time

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DSTALGO

```
namespace FindName
{
    public partial class Form1 : Form

    {
        string[] names = { "Ana", "Ben", "Carlo", "David" };
        public Form1()
        {
            InitializeComponent();
        }

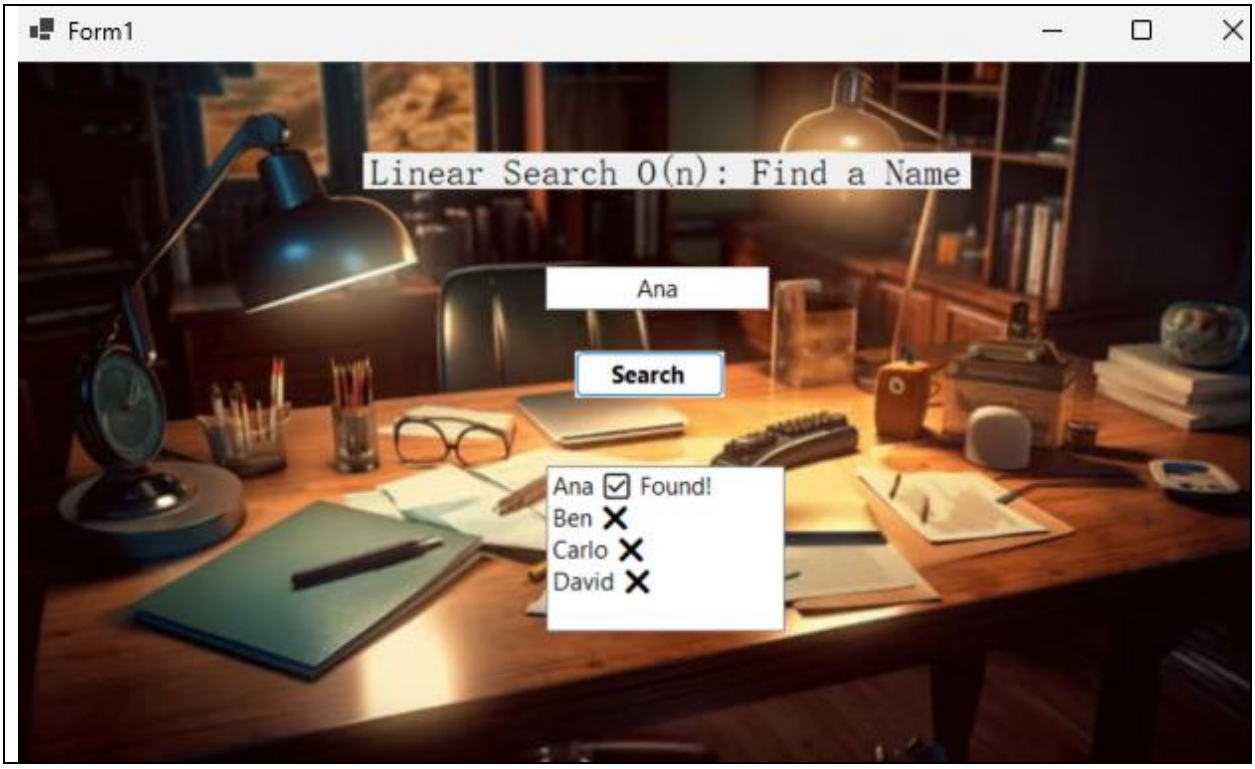
        private void btnSearch_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();
            string target = txtSearch.Text.Trim();
            bool found = false;

            // Linear search O(n): check each name
            foreach (string name in names)
            {
                if (name == target)
                {
                    lstOutput.Items.Add($"{name} ☑ Found!");
                    found = true;
                }
                else
                {
                    lstOutput.Items.Add($"{name} ❌");
                }
            }

            if (!found)
            {
                MessageBox.Show("Name not found.", "Result");
            }
        }
    }
}
```

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DSTALGO



3rd Linear Time

```
namespace CalculateTotal
{
    public partial class Form1 : Form
    {
        int[] numbers = { 5, 10, 15 };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnCalculate_Click(object sender, EventArgs e)
        {
            lstNumbers.Items.Clear();
            int total = 0;

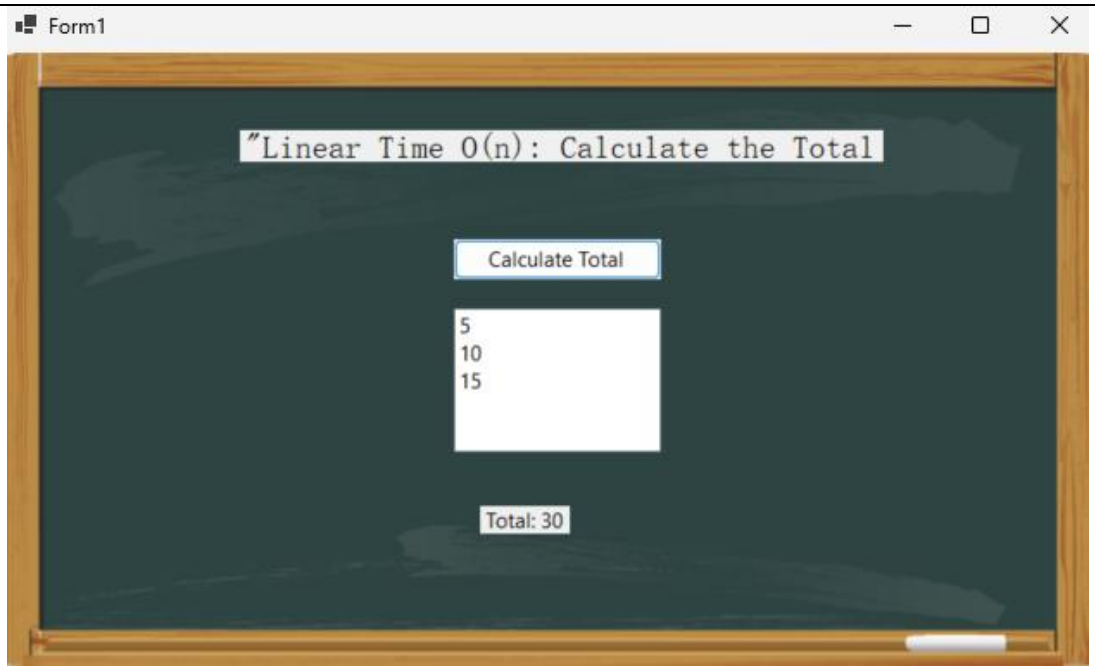
            // Linear time O(n): add each number
            foreach (int number in numbers)
            {
                lstNumbers.Items.Add(number);
                total = total + number;
            }

            lblTotal.Text = $"Total: {total}";
```

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DSTALGO

```
        }
    }
}
```



1st Logarithmic Time

```
namespace BinarySearch
{
    public partial class Form1 : Form
    {
        int[] numbers = { 10, 20, 30, 40, 50, 60, 70 };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnSearch_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();
            if (!int.TryParse(txtSearch.Text.Trim(), out int target))
            {
                MessageBox.Show("Please enter a valid number.", "Error");
                return;
            }
```

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```
            int left = 0;
            int right = numbers.Length - 1;
            bool found = false;

            // Binary Search O(log n)
            while (left <= right)
            {
                int middle = (left + right) / 2;
                lstOutput.Items.Add($"Checking index {middle}: {numbers[middle]}");

                if (numbers[middle] == target)
                {
                    lstOutput.Items.Add($"{target} ☑ Found!");
                    found = true;
                    break;
                }
                else if (numbers[middle] < target)
                {
                    lstOutput.Items.Add($"{numbers[middle]} < {target} → Search
Right");
                    left = middle + 1;
                }
                else
                {
                    lstOutput.Items.Add($"{numbers[middle]} > {target} → Search
Left");
                    right = middle - 1;
                }
            }

            if (!found)
            {
                MessageBox.Show("Number not found.", "Result");
            }
        }
    }
}
```

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DSTALGO



2nd Logarithmic Time

```
namespace Divide2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnDivide_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();

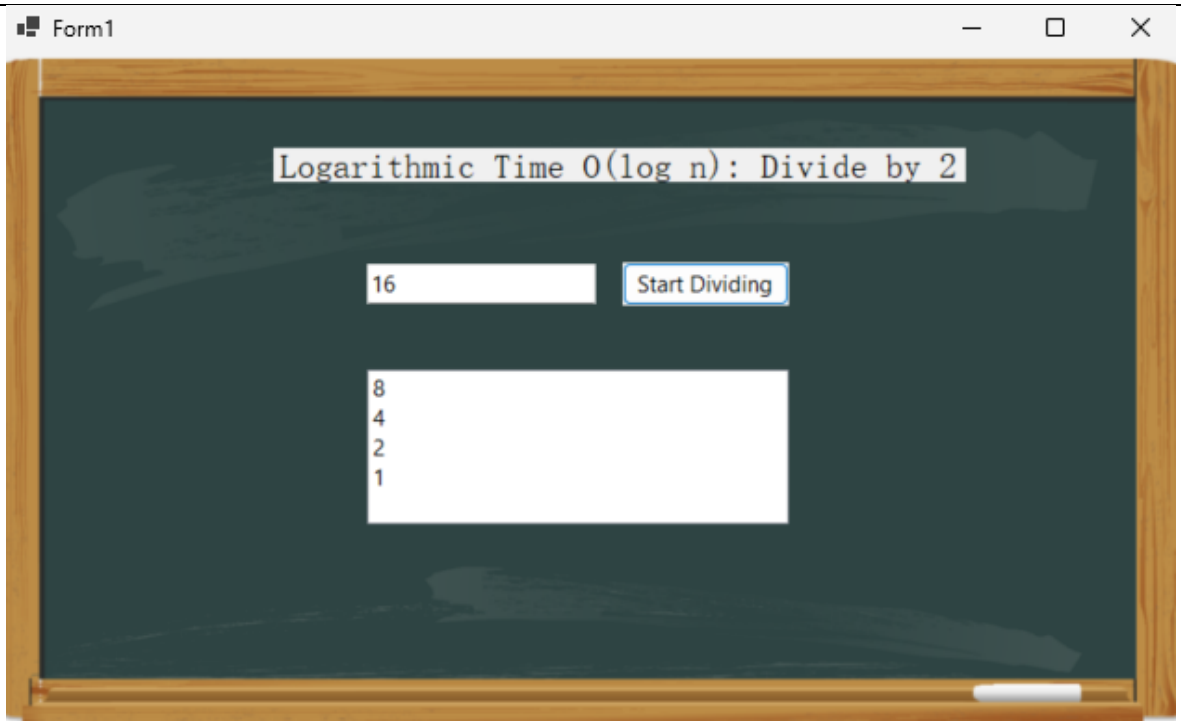
            if (!int.TryParse(txtNumber.Text.Trim(), out int n) || n <= 1)
            {
                MessageBox.Show("Please enter a number greater than 1.", "Error");
                return;
            }

            // Logarithmic time O(log n): keep dividing by 2
            while (n > 1)
            {
                n = n / 2;
```

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DSTALGO

```
                lstOutput.Items.Add(n);
            }
        }
    }
}
```



3rd Logarithmic Time

```
namespace BinaryGuessing
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnGuess_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();
```

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DSTALGO

```
            if (!int.TryParse(txtHigh.Text.Trim(), out int high) || high < 1)
            {
                MessageBox.Show("Please enter a valid number greater than 0.",
"Error");
                return;
            }

            int low = 1;

            // Binary-search-style guessing O(log n)
            while (low <= high)
            {
                int middle = (low + high) / 2;
                lstOutput.Items.Add($"Guess: {middle}");

                high = middle - 1;
            }
        }
    }
}
```



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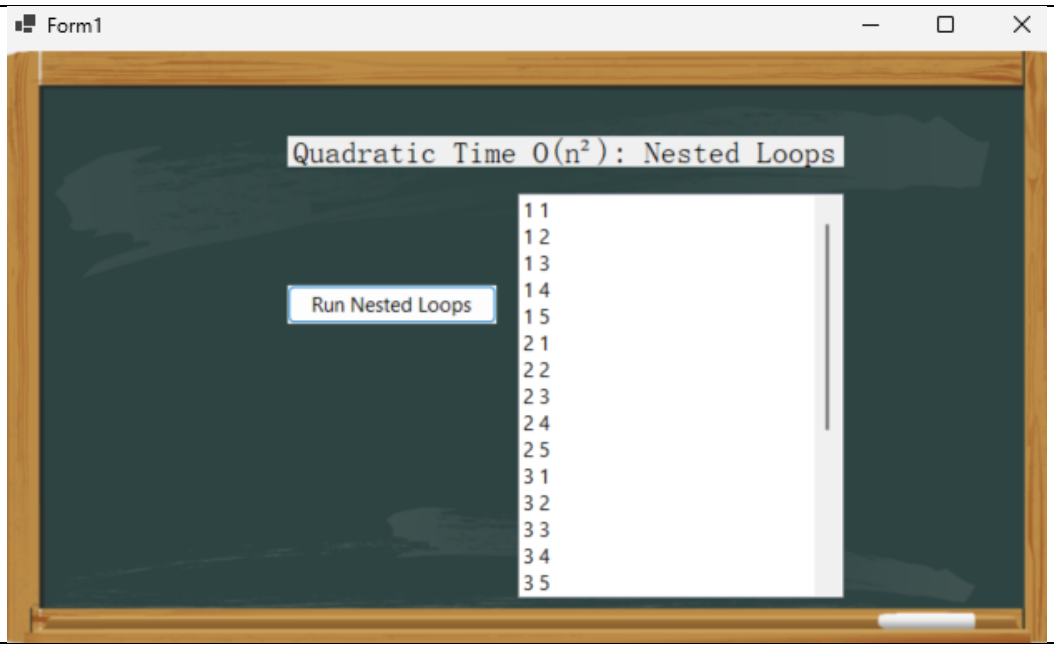
DSTALGO

1st Quadratic Time

```
namespace QuadraticTime01
{
    public partial class Form1 : Form
    {
        int[] numbers = { 1, 2, 3, 4, 5 };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnRun_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();

            // Quadratic time O(n²): nested loops
            for (int i = 0; i < numbers.Length; i++)
            {
                for (int j = 0; j < numbers.Length; j++)
                {
                    lstOutput.Items.Add($"{numbers[i]} {numbers[j]}");
                }
            }
        }
    }
}
```



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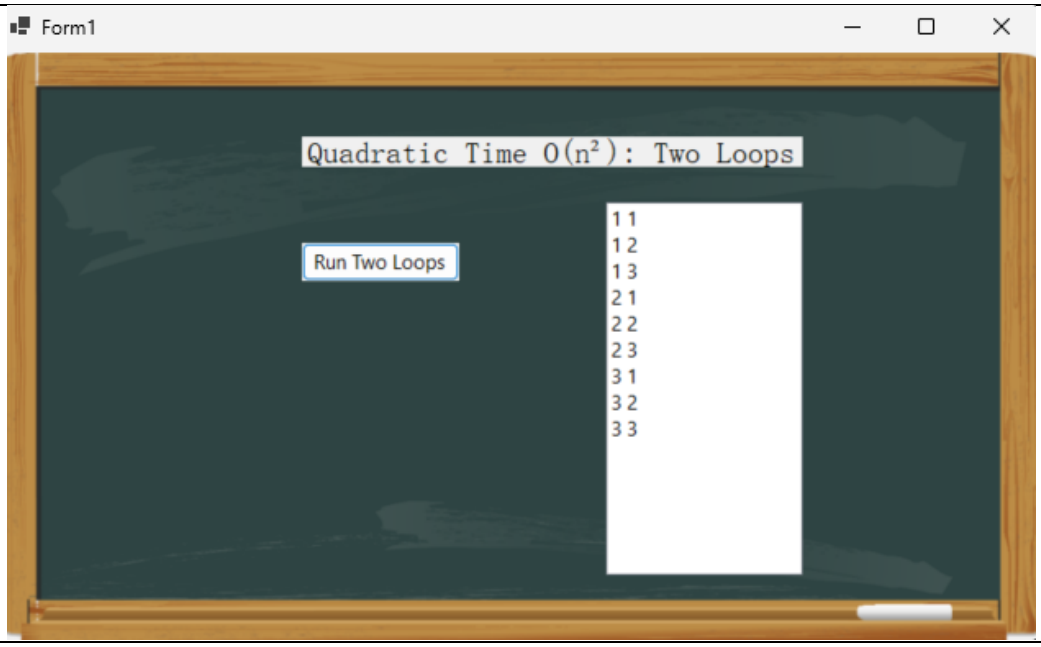
DSTALGO

2nd Quadratic Time

```
namespace TwoLoops
{
    public partial class Form1 : Form
    {
        int[] numbers = { 1, 2, 3 };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnRun_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();

            // Quadratic time O(n²): nested loops
            for (int i = 0; i < numbers.Length; i++)
            {
                for (int j = 0; j < numbers.Length; j++)
                {
                    lstOutput.Items.Add($"{numbers[i]} {numbers[j]}");
                }
            }
        }
    }
}
```



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3rd Quadratic Time

```
namespace CompareStudents
{
    public partial class Form1 : Form
    {
        string[] students = { "Ana", "Ben", "Carlo" };
        public Form1()
        {
            InitializeComponent();
        }

        private void btnCompare_Click(object sender, EventArgs e)
        {
            lstOutput.Items.Clear();

            // Quadratic time O(n²): nested loops
            for (int i = 0; i < students.Length; i++)
            {
                for (int j = 0; j < students.Length; j++)
                {
                    lstOutput.Items.Add($"{students[i]} checks {students[j]}");
                }
            }
        }
    }
}
```



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DSTALGO

1st Exponential Time

```
namespace LightSwitch
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnCalculate_Click(object sender, EventArgs e)

        {
            if (!int.TryParse(txtSwitches.Text.Trim(), out int n) || n < 1)
            {
                MessageBox.Show("Please enter a valid number of switches (greater than
0).", "Error");
                return;
            }

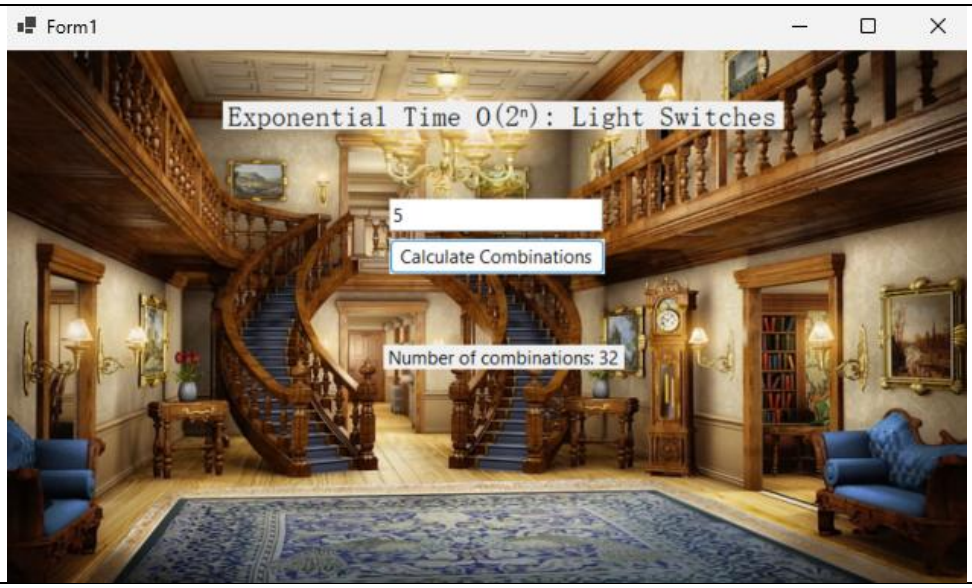
            int combinations = 1;

            // Exponential time O(2^n): each switch doubles the combinations
            for (int i = 0; i < n; i++)
            {
                combinations = combinations * 2;
            }

            lblResult.Text = $"Number of combinations: {combinations}";
        }
    }
}
```

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DSTALGO



2nd Exponential Time

```
namespace CoinToss
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnCalculate_Click(object sender, EventArgs e)
        {
            if (!int.TryParse(txtTosses.Text.Trim(), out int tosses) || tosses < 1)
            {
                MessageBox.Show("Please enter a valid number of tosses (greater than
0).", "Error");
                return;
            }

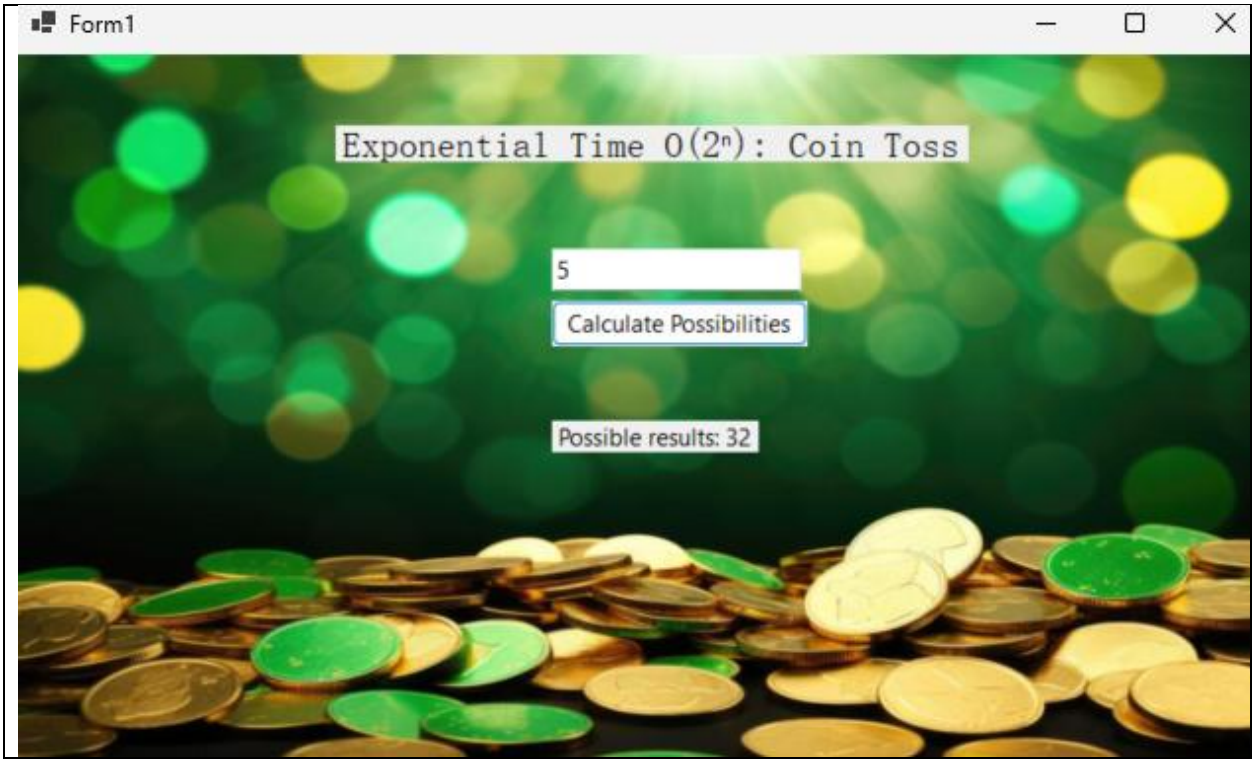
            int possibilities = 1;

            // Exponential time O(2^n): each toss doubles the possibilities
            for (int i = 0; i < tosses; i++)
            {
                possibilities *= 2;
            }

            lblResult.Text = $"Possible results: {possibilities}";
        }
    }
}
```

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DSTALGO



3rd Exponential Time

```
namespace PassChoice
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }
        private void btnCalculate_Click(object sender, EventArgs e)
        {
            lstPasswords.Items.Clear();

            if (!int.TryParse(txtPositions.Text.Trim(), out int positions) ||
positions < 1)
            {
                MessageBox.Show("Please enter a valid number of positions (greater
than 0).", "Error");
                return;
            }
```

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DSTALGO

```
            int possibilities = 1;

            // Exponential time O(2ⁿ): each position doubles the possibilities
            for (int i = 0; i < positions; i++)
            {
                possibilities *= 2;
            }

            lblResult.Text = $"Possible passwords: {possibilities}";

            // Generate all possible passwords using A and B
            GeneratePasswords(positions, "");
        }

        private void GeneratePasswords(int positions, string current)
        {
            if (current.Length == positions)
            {
                lstPasswords.Items.Add(current);
                return;
            }

            // Recursive branching: each position can be A or B
            GeneratePasswords(positions, current + "A");
            GeneratePasswords(positions, current + "B");
        }
    }
}
```

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